



CVL2001: Climate Change and Adaptation

Lecture 3: Impacts of Climate Change

3.1 Why Does a Few Degrees of Warming Matter?

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Aug 10, 2026

Section B

Today's Outline

- Why warming matters
- Temperature Anomaly
- Precipitation Anomaly
- Sea-level rise

The Summer That Changed the Conversation

BEFORE SUMMER 2010 — VLADIMIR PUTIN

- “Russia is a northern country. Warmer temperatures might not be so bad—we could spend less on warm coats.”



AFTER SUMMER 2010 — DMITRI MEDVEDEV

“Practically everything is burning. The weather is anomalously hot.” The crisis should be a wake-up call for stronger action on climate change.



What changed between these two statements?

Q) We know that the planet is warming. Can you think of any places that might initially benefit from global warming?

- Which places could experience milder winters?
- What benefits might they receive?
- Would those benefits continue as warming increases?

Munnar: Pleasant for Tourists, Risky for Tea?

- Tea is cultivated around 1,600 m elevation
- 30 years of data: 1996–2025
- Summer and winter maximum temperatures increased
- June rainfall declined and monsoon timing shifted
- Morning humidity declined
- A small climatic shift can disrupt a crop adapted to Munnar's high-range conditions.



Image source: <https://www.keralaruchi.com/places/munnar>

The Himalayas Are Changing Before Our Eyes

- Ghepang Ghat glacial lake, Himachal Pradesh
- 1989: 36.49 hectares
- 2022: 101.30 hectares
- Increase: 178%
- As glaciers retreat, lakes expand, and downstream flood risk can increase. This is called a glacial lake outburst flood, or GLOF.

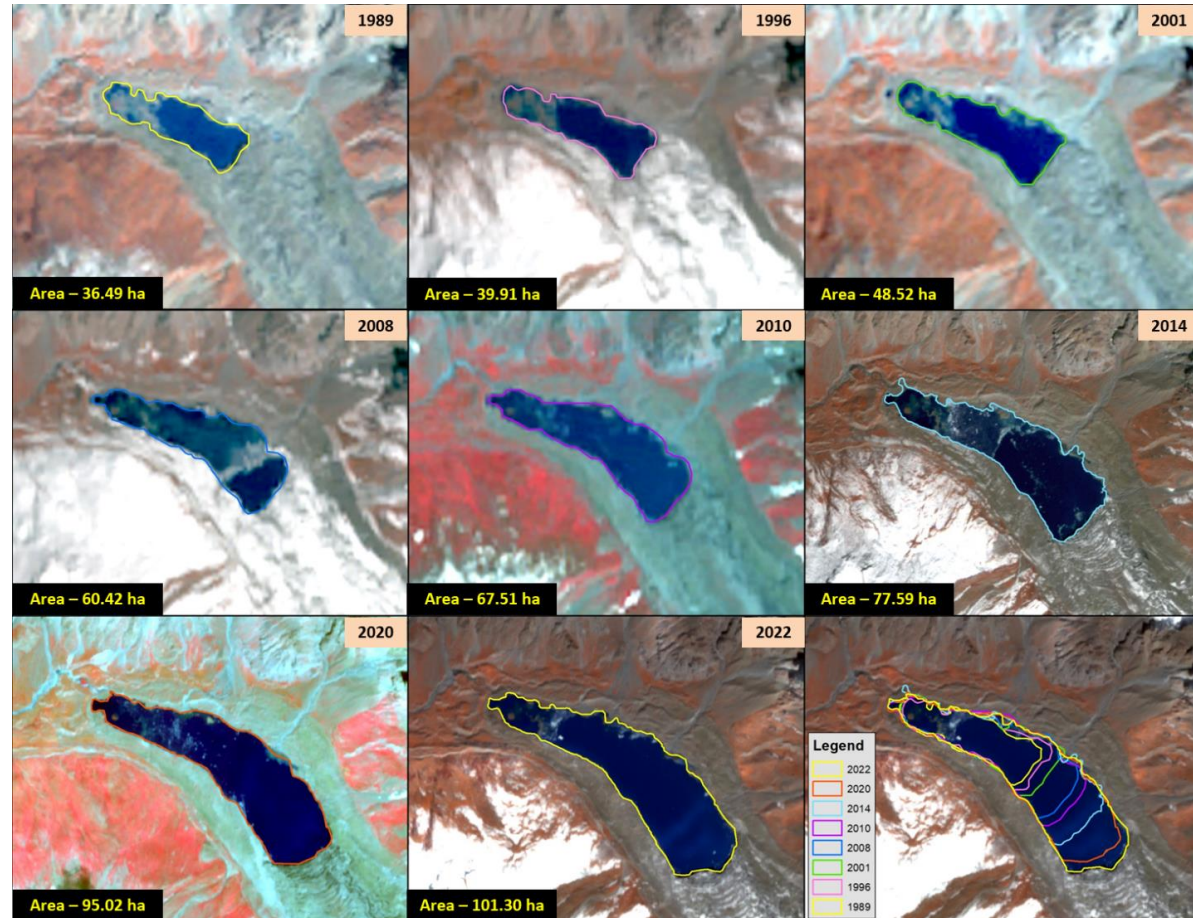


Image source: <https://www.drishtiias.com/daily-updates/daily-news-analysis/expanding-glacial-lakes-in-the-himalayas>

A Few Degrees Globally Is Not a Small Change

- Local weather can vary by many degrees in a single day.
- Global mean temperature averages the entire planet over the whole year.
- Raising this average requires an enormous accumulation of heat.
- Some regions and seasons will warm much more than the global average.

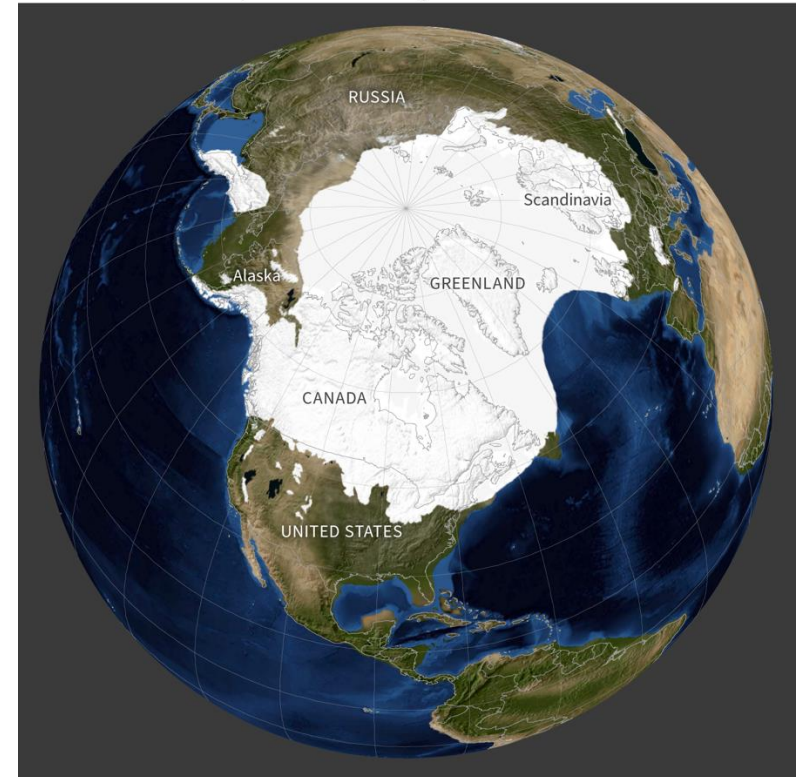
A 2°C warmer planet is not the same as a 2°C warmer afternoon.

Six Degrees Separated Two Different Worlds

- During the last ice age, global temperature was only about 6°C colder than today.
- Vast ice sheets covered much of North America and Europe.
- Global sea level was approximately 120 metres lower.
- Coastlines, ecosystems and rainfall patterns were dramatically different.

A few degrees of global change can transform the entire planet.

Ice sheet extent near the peak of the last ice age



19,000 BC

NOAA Climate.gov
Data: NOAA SOS

Global Means Hide Larger Regional Changes

- Land usually warms more than ocean.
- High latitudes can warm faster than the global average.
- Heat extremes shift as the average climate moves.
- Rainfall and circulation changes create additional impacts.
- The global number is the starting point, not the local forecast.

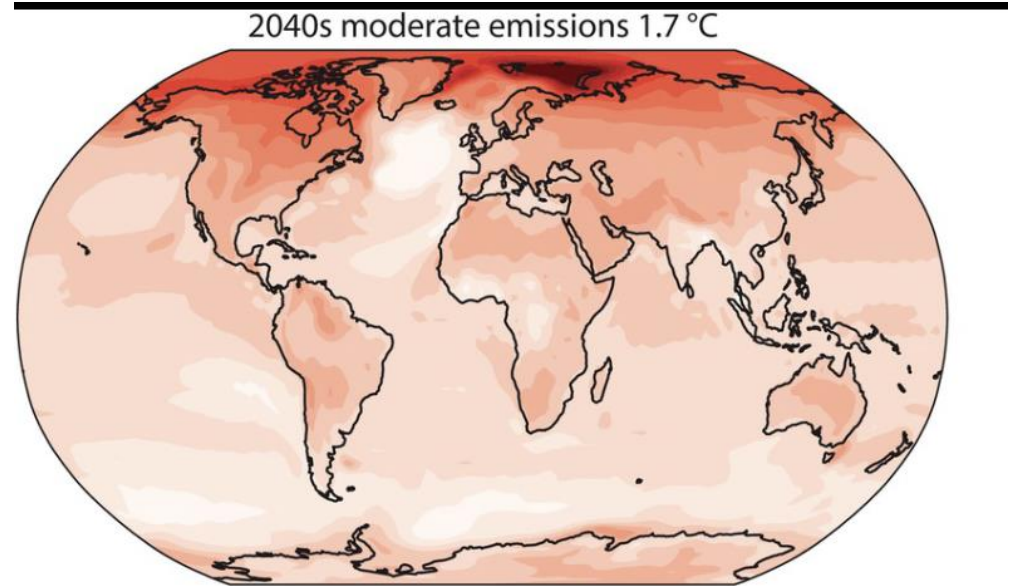


Image source : Dessler, A. E., Introduction to Modern Climate Change, Chapter 9.

But Climate Has Changed Naturally Before

- Earth's climate has changed because of orbital cycles, volcanic eruptions and variations in solar energy.
- These past changes confirm that climate responds when its energy balance is disturbed.
- What matters is the cause, magnitude and rate of the present change.
- Rapid warming gives societies and ecosystems less time to adjust.

Natural climate change in the past does not make today's rapid warming harmless.

Rate Determines How Much Time We Have

- Slow change gives societies and ecosystems more time to adjust.
- Rapid warming can outpace crop improvement and species migration.
- Buildings, drainage systems and coastal development last for decades.
- Today's decisions can therefore lock in future risks.

The faster the change, the more difficult and costly adaptation becomes.



Image source: <https://ychef.files.bbci.co.uk/1280x720/p0d1xyd2.jpg>

Why 2026 European Heatwave was Deadly

- Risk depends on temperature and preparedness.
- Homes may be designed to retain heat, not release it.
- Air-conditioning access can be limited.
- Health systems face a sudden surge in demand.
- Older people and outdoor workers are especially exposed.
- The same heat produces different impacts in different places.

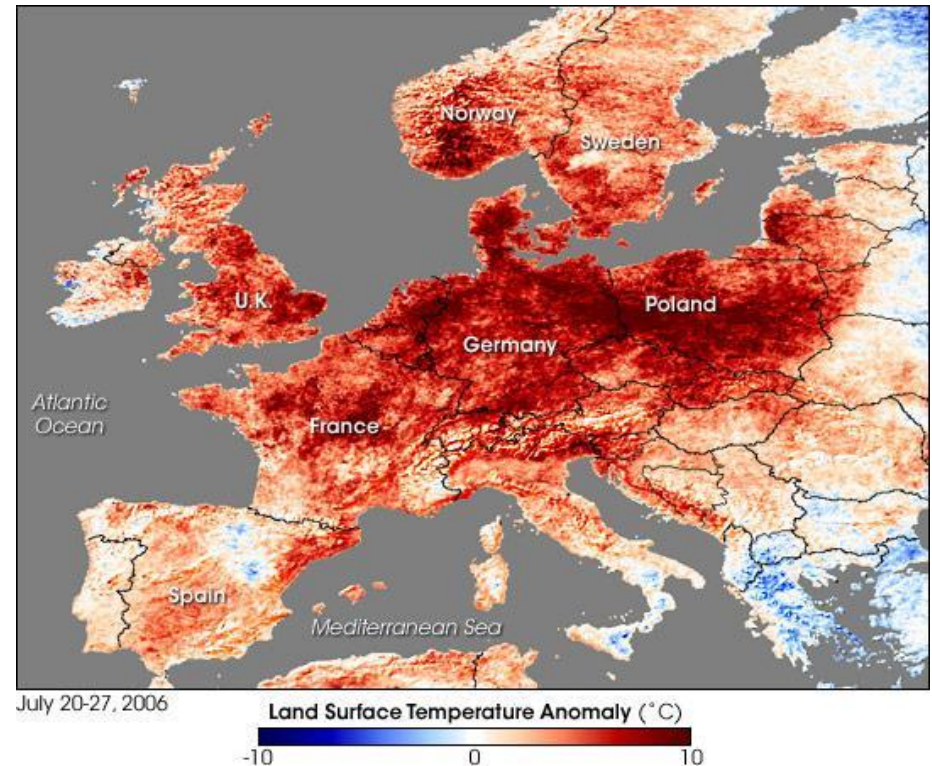


Image source:
https://europe.noharm.org/sites/default/files/styles/max_1300x1300/public/content-images/europe/5540/2018-08-23%20heatwave%20in%20Europe.jpg?itok=mMresghQ

1. Temperature Anomaly

- Compare the moderate- and high-emission pathways.
- In the **2040s**, the projected warming remains relatively similar.
- By the **2090s**, the pathways diverge substantially.
- Land and high-latitude regions warm more than the global average.

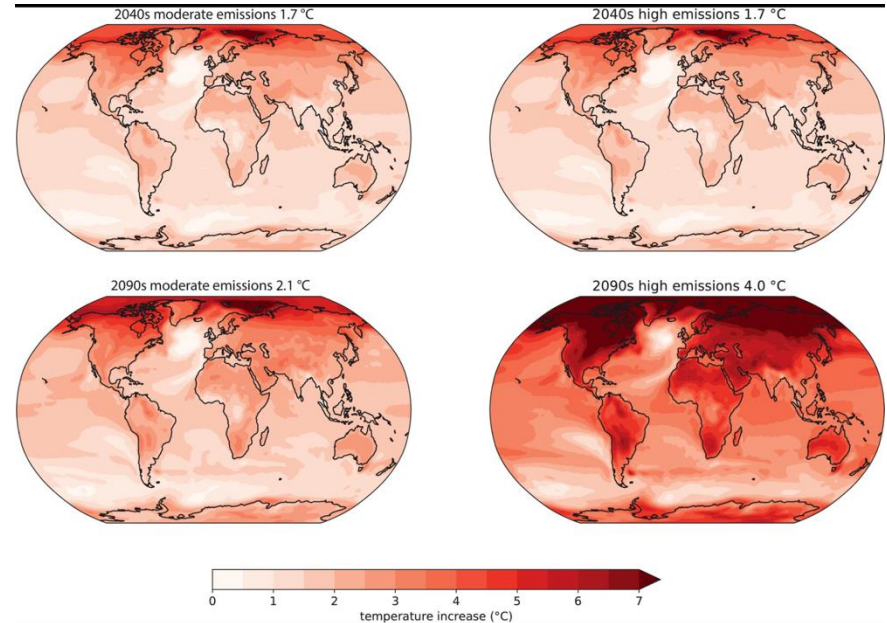


Image source : Dessler, A. E., Introduction to Modern Climate Change, Chapter 9.

Today's emissions choices have their greatest influence later this century.

1. Temperature Anomaly...

- In the 2040s, the emission pathways remain relatively close.
- By the 2090s, their projected warming differs substantially.
- Present emissions therefore determine the scale of late-century impacts.
- Some adaptation is already necessary, while mitigation limits how much more will be required.

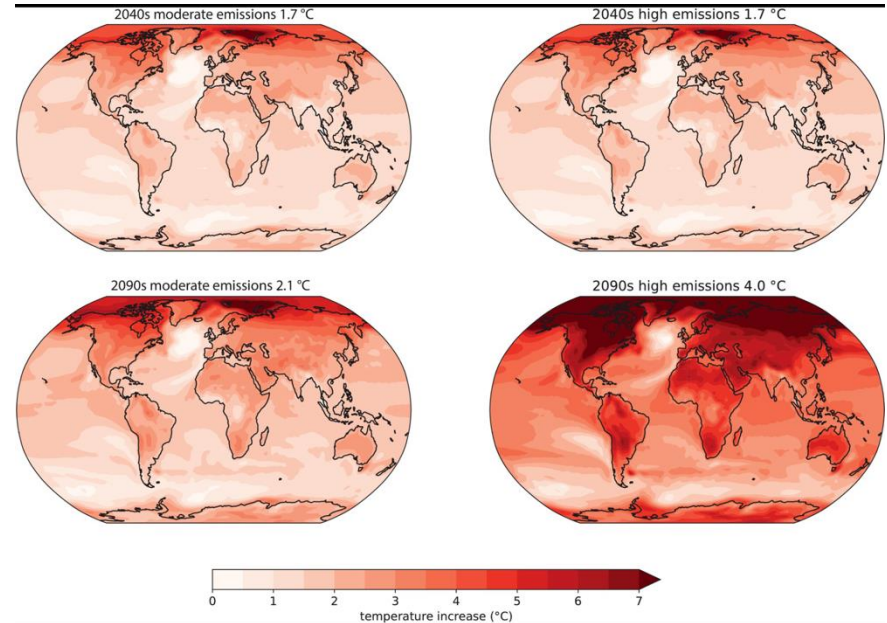


Image source : Dessler, A. E., Introduction to Modern Climate Change, Chapter 9.

**Mitigation determines the magnitude of change;
adaptation determines how we manage it.**

2. Precipitation Anomaly

- Warmer air can hold more water vapour.
- Higher temperatures increase evaporation where water is available.
- More atmospheric moisture can intensify heavy rainfall.
- Other regions may experience greater drying as circulation patterns change.

“Wet gets wetter and dry gets drier” is a useful global tendency—not a rule for every location.

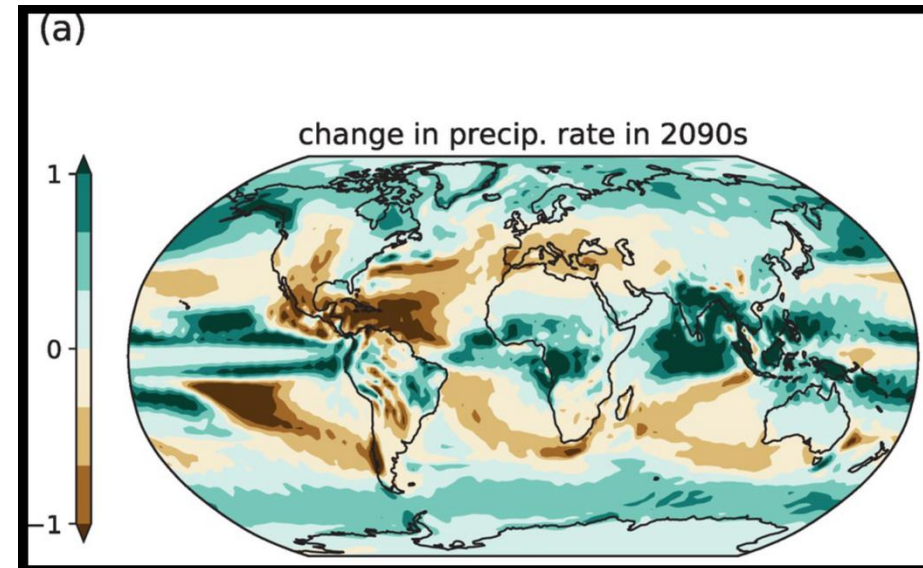


Image source : Dessler, A. E., Introduction to Modern Climate Change, Chapter 9.

2. Precipitation Anomaly...

- Heavier rainfall can increase flash floods, river flooding and drainage failure.
- Longer dry periods can reduce soil moisture, stress crops and intensify water scarcity.
- The same region may experience both extremes at different times.
- **Local impacts depend on rainfall timing, intensity and preparedness.**

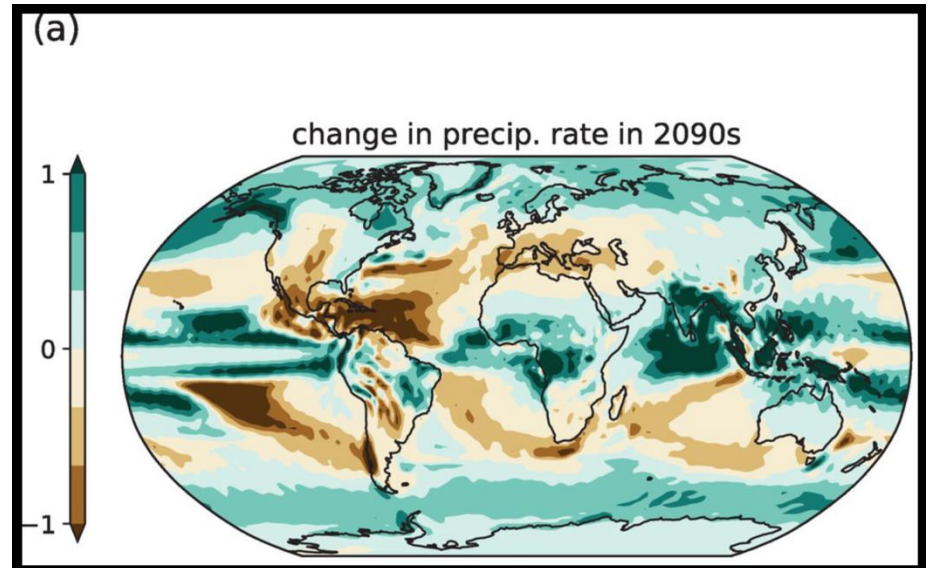


Image source : Dessler, A. E., Introduction to Modern Climate Change, Chapter 9.

3. Sea Level Rise

Why Does Sea Level Rise?

- **Melting grounded ice:** Meltwater from glaciers and ice sheets enters the ocean, increasing its total amount of water.
- **Thermal expansion:** Water expands as it warms, increasing the volume of the ocean.
- Measurements confirm that sea level has risen as global temperatures have increased.
- Sea level is projected to rise 47–73 cm above 1995–2014 levels by 2100, compared with about 18 cm during the twentieth century.
- Sea-level rise is a certain and long-lasting impact of climate change.



Image source: <https://ychef.files.bbci.co.uk/1280x720/p0d1xyd2.jpg>

3.Sea Level Rise

Q) What Happens Before the Coast Is Submerged?

A higher mean sea level raises the starting point for:

- Storm surges
- High-tide flooding
- Coastal erosion
- Saltwater intrusion into groundwater

Which of these impacts can occur long before permanent inundation?

Alappuzha, Kerala: Water from Both Sides

- Low-lying land lies between the sea and the backwaters.
- Higher sea level can obstruct drainage and worsen coastal flooding.
- Saltwater can enter groundwater, soils and agricultural areas.
- Heavy rainfall and high tides can occur together.
- **Sea-level rise can affect communities before permanent inundation occurs.**



Video Source:
<https://www.youtube.com/shorts/wW75v1qGk4s?feature=share>

Mumbai: Multiple Hazards, One Rising Baseline

- Mumbai concentrates people, infrastructure and economic activity near the coast.
- Monsoon rainfall already creates severe urban flooding.
- High tides can restrict the drainage of rainwater into the sea.
- Sea-level rise raises the baseline for tides, storm surges and coastal flooding.
- **Climate change magnifies risks that the city already faces.**



Image source: https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcTyhLiwCvrfEwI8ydbQeW_CfzqzD4AweBFW4pgwi3Rjtv968KuvnTVFyo&s=10

Key Takeaways

- A few degrees of global warming can transform climate systems and coastlines.
- Impacts depend on both the magnitude and the rate of change.
- Warming intensifies heat, rainfall extremes, drought and sea-level rise.
- Risks are greatest where people, ecosystems and infrastructure are highly exposed and vulnerable.
- Mitigation limits future change; adaptation reduces the harm we cannot avoid.

Climate change is global, but its impacts—and our responses—are local.